



Lecture 19:

VLSI High-Speed I/O Architectures



This Topic: VLSI High-Speed I/O Architectures

- Introduction
- Common Clock Signaling I/O Circuits
- Forward Clock Signaling I/O Circuits
- Embedded Clock Signaling I/O Circuits

Introduction



- VLSI I/O architecture have been going through three basic generations:
 - Common Clock (or Synchronous Clock) signaling
 - Forward Clock (or Source Synchronous Clock) signaling
 - Embedded Clock (or Clock Recovery) signaling
- Currently all three types I/O circuits are seen in practical applications
- Some high-speed I/O circuits combine multiple schemes for better performance

Common Clock I/O Architecture

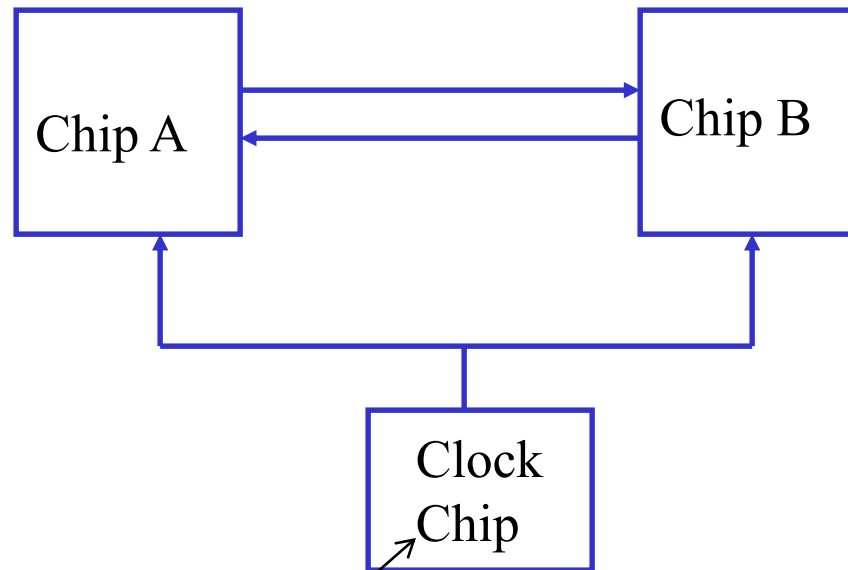


- Common clock I/O is also known as synchronous I/O
- Very popular interface type used in traditional computer systems
- Common clock I/O was a “natural” migration from on-chip circuits
- Common clock distribution was a challenge for high-speed I/O applications
 - Maintain small clock skew among all clocked blocks in system
 - Data rates are typically limited to around 100Mb/s

Common Clock I/O Operation

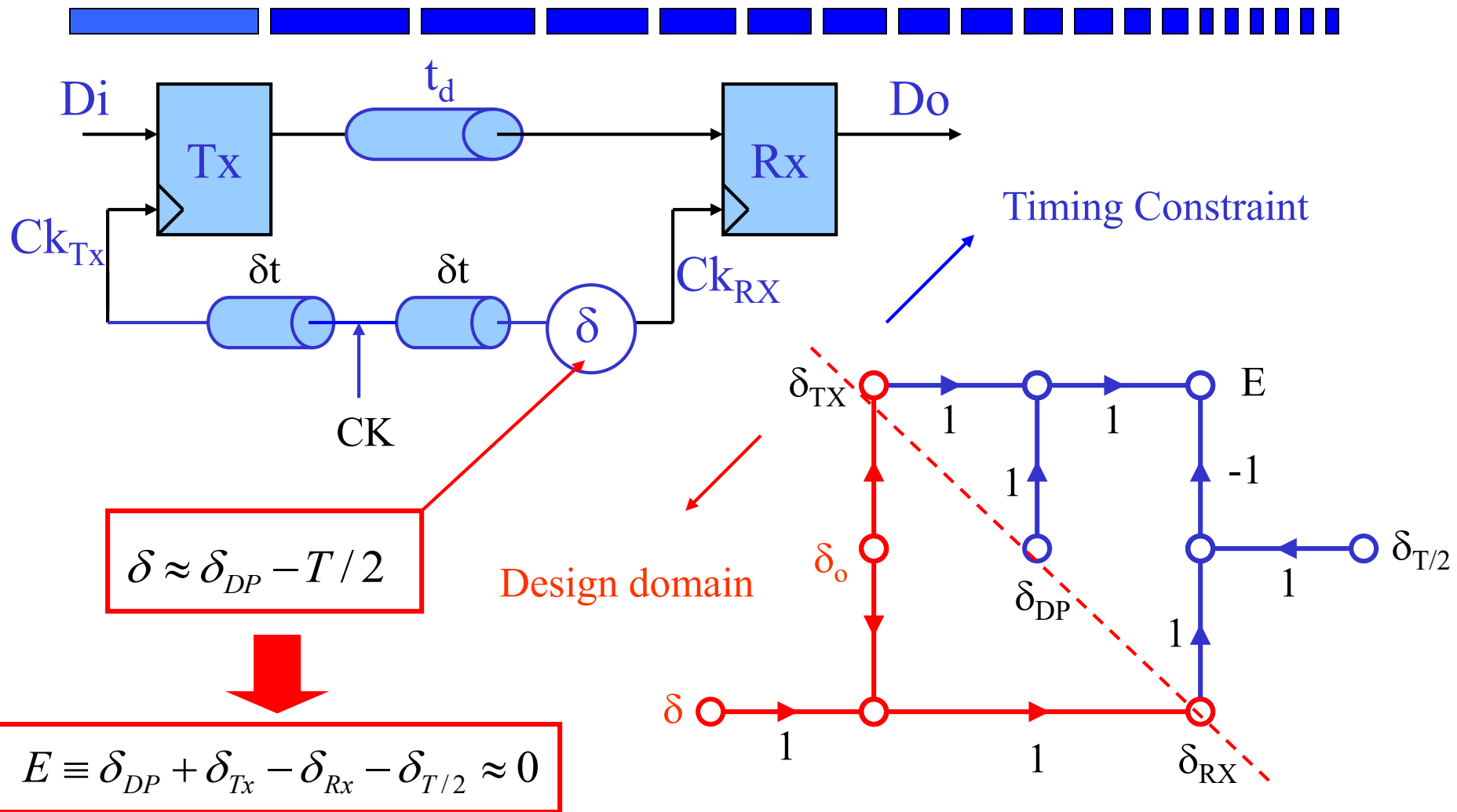


- Tx and Rx supplied by a common clock source
- Very popular in old computer I/O generations



Centralized Source Clock

Common Clock I/O Circuits

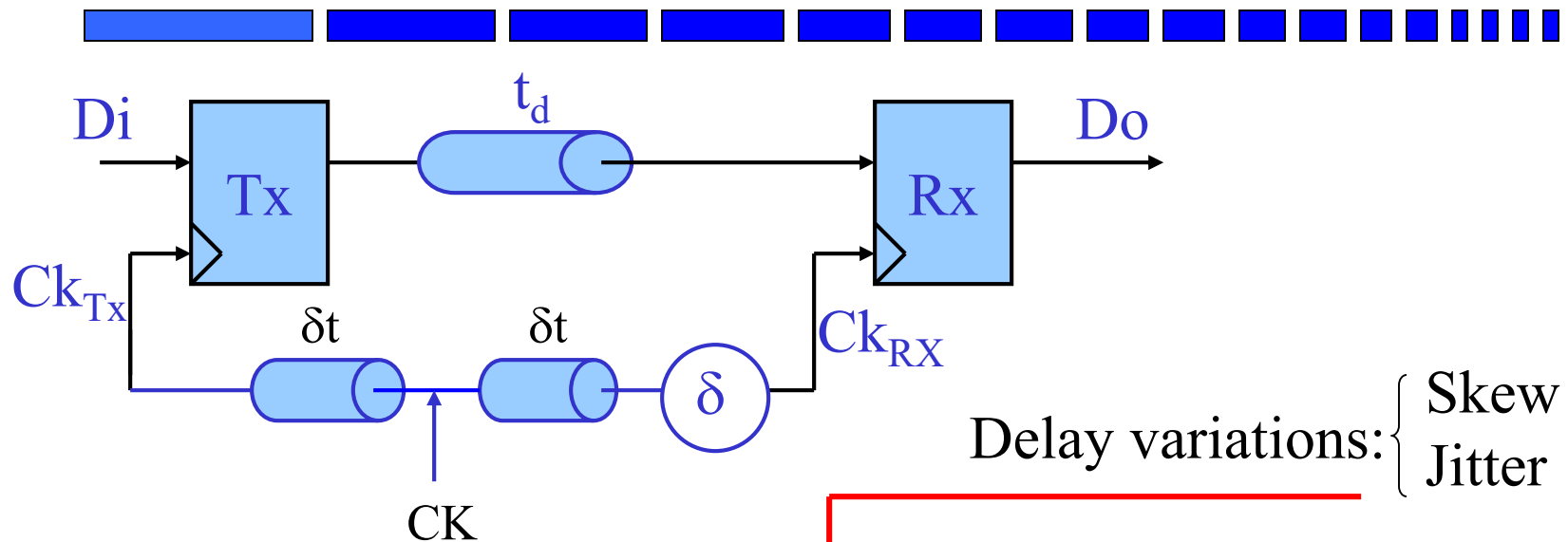


Limitation of Common Clock I/O Circuits



- Major challenges of practical synchronous clocking I/O design are to control the clock skews and the propagation delay for both
 - Absolute values and
 - Variations
- Since these delay elements have very low correlations, no skew/jitter compensation effect can be used
 - Popular for on-chip circuit
 - Limited speed on I/O applications

Common Clock I/O Circuit Data Rate



$$E \equiv \Delta(\delta_{DP} + \delta_{Tx} - \delta_{Rx} - \delta_{T/2}) \leq E_{Max} \equiv \frac{T - (t_{setup} + t_{hold})}{2}$$

$$T > 2[\Delta(\delta_{DP} + \delta_{Tx} - \delta_{Rx} - \delta_{T/2}) + (t_{setup} + t_{hold})]$$

(1~10) ns

Example: Common Clock I/O Circuits



- 100 MHz bus
 - Minimum clock period = 10 ns
- Given:
 - Maximum skew = 250 ps
 - Maximum edge-edge jitter = 250 ps.
- Calculate the minimum effective clock period:
 - minimum effective period =
 - minimum period – maximum skew – maximum jitter
 - min effective period = 10.0 ns – 0.25 ns – 0.25 ns = 9.5 ns
- Therefore, maximum allowed for silicon plus interconnect delay is 9.5 ns.

Forward Clock I/O Architecture



- Forward clock is also known as the source synchronous clock
- Forward clock scheme has been used as an enhancement to compensate for clock distribution skews
- Forward clock I/O is a “matched-delay” based clocking scheme
- Significantly improved timing control with data rate $> 1 \text{ Gb/s}$
- Application examples: CSI, DDR, Mipi D-PHY Interface

Forwarded Clock I/O Operation



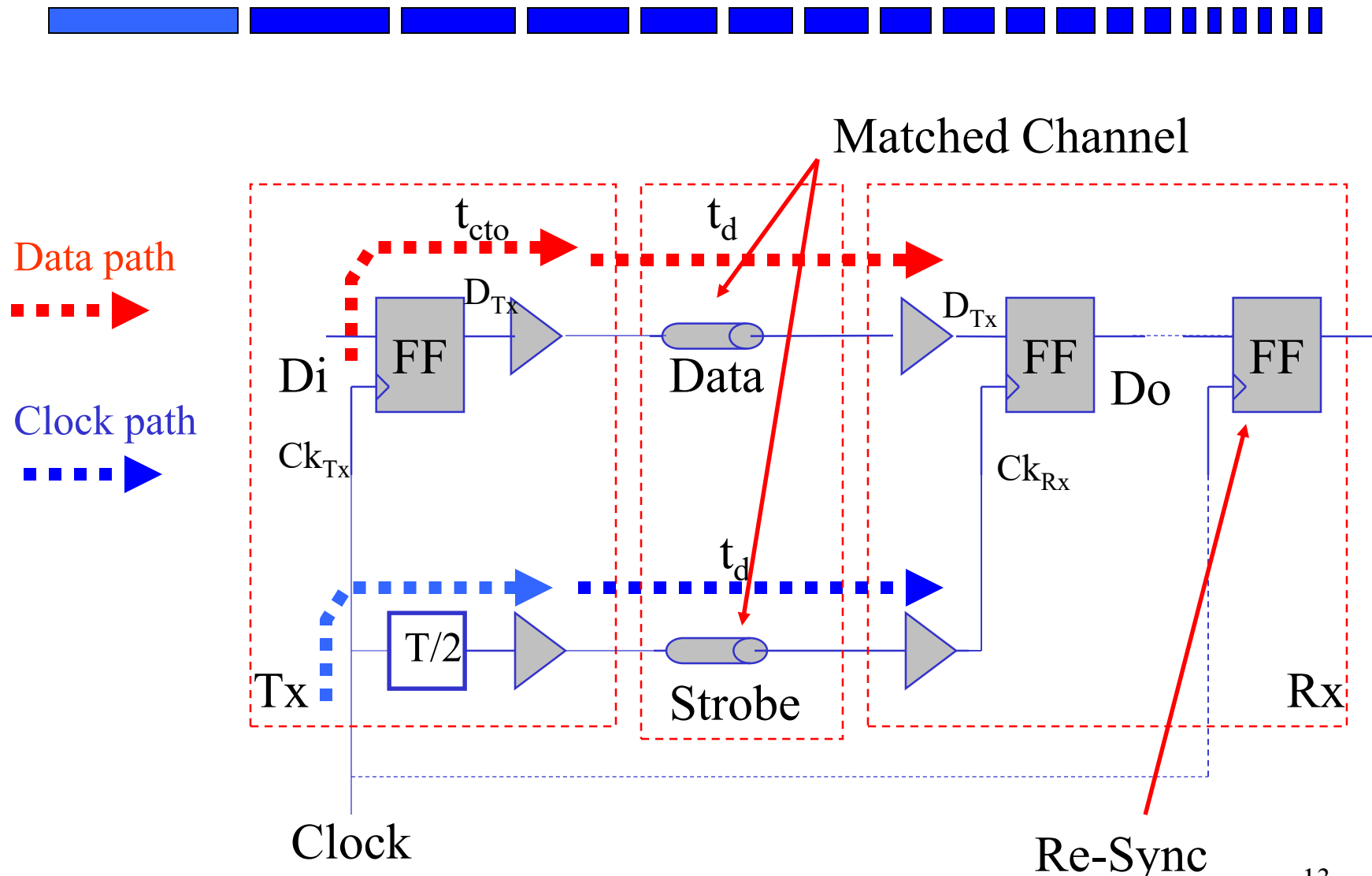
- Forward clocking I/O send both data and clock (strobe) from the Tx using matched channel
 - Fixed phase relation between data and strobe
- Design concept shift from absolute value based to delay-matching based design
 - Higher performance
 - Longer allowable signal/clock routing distance

Forwarded Clock I/O Operation

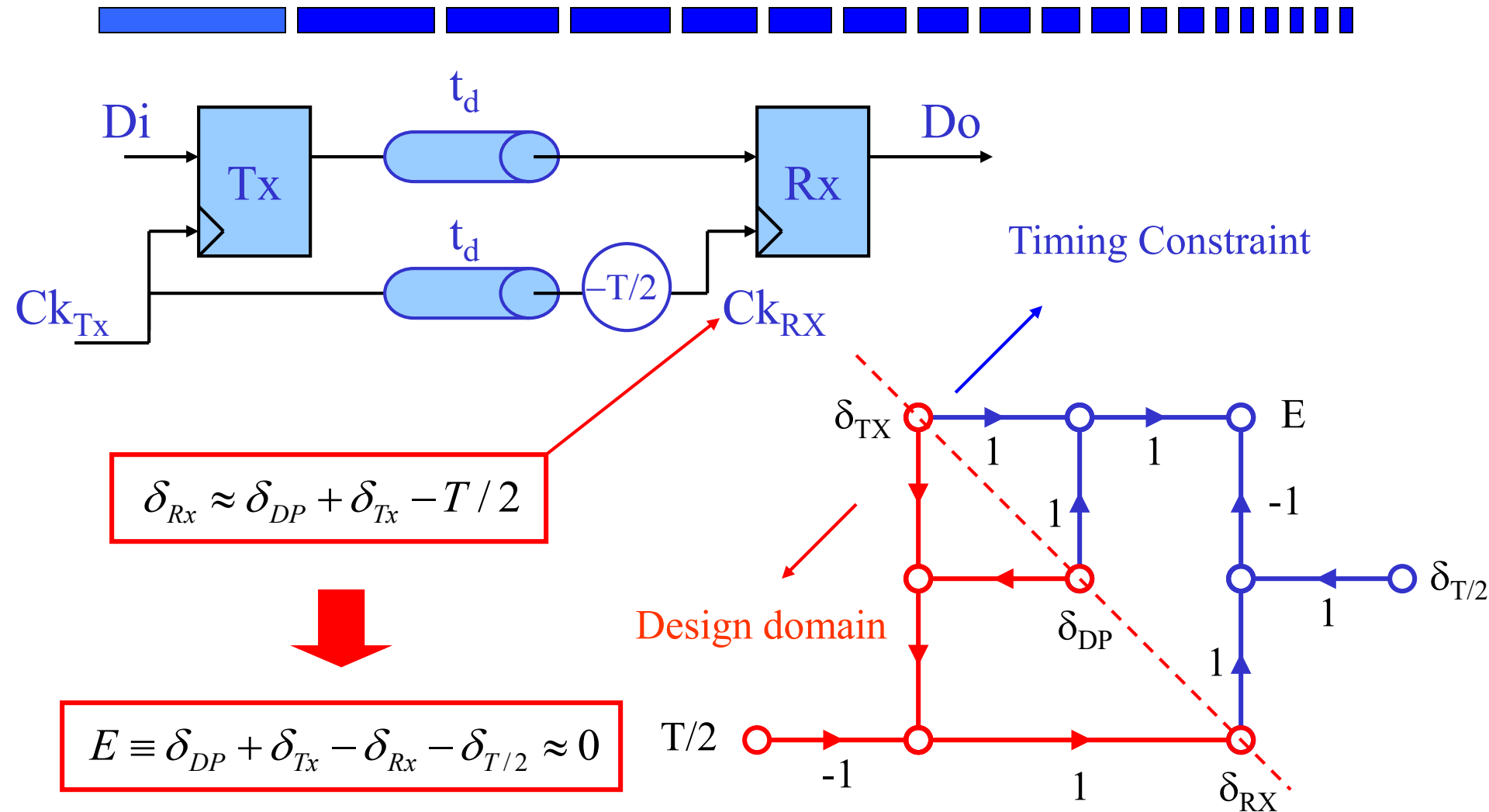


- Data D_{Tx} generated the Tx synced to the Tx clock rise edges
- The Tx clock is send through the MATCHED channel as the data to generate the Rx clock.
- Data D_{Rx} in Rx is latched using the received clock from Tx
- Data D_o is re-synced to local Rx clock

Forwarded Clocking I/O Operation



Forward Clock I/O Circuits



Common Sources of Clock Skew



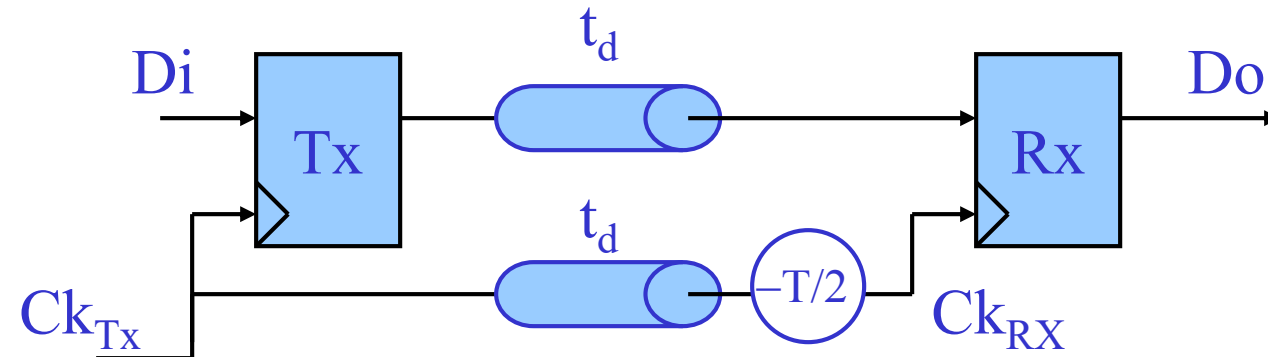
- Drive strength mismatch in clock buffers
- Loading mismatch of the clock buffers
- Length mismatch of Interconnect
- Jitter amplification effects
- Duty-Cycle Variations

Limitation of Forward Clocking I/Os



- Since these delay elements of the data and strobe has high correlations and skew/jitter compensation effect can now be used
 - Significant improvement in data rate
- Jitter amplification and clock de-skewing are key design considerations

Forward Clock I/O Circuit Data Rate



Delay variations: { Skew (mismatch)
Jitter

$$E \equiv \Delta(\delta_{DP} + \delta_{Tx} - \delta_{Rx} - \delta_{T/2}) \leq E_{Max} \equiv \frac{T - (t_{setup} + t_{hold})}{2}$$

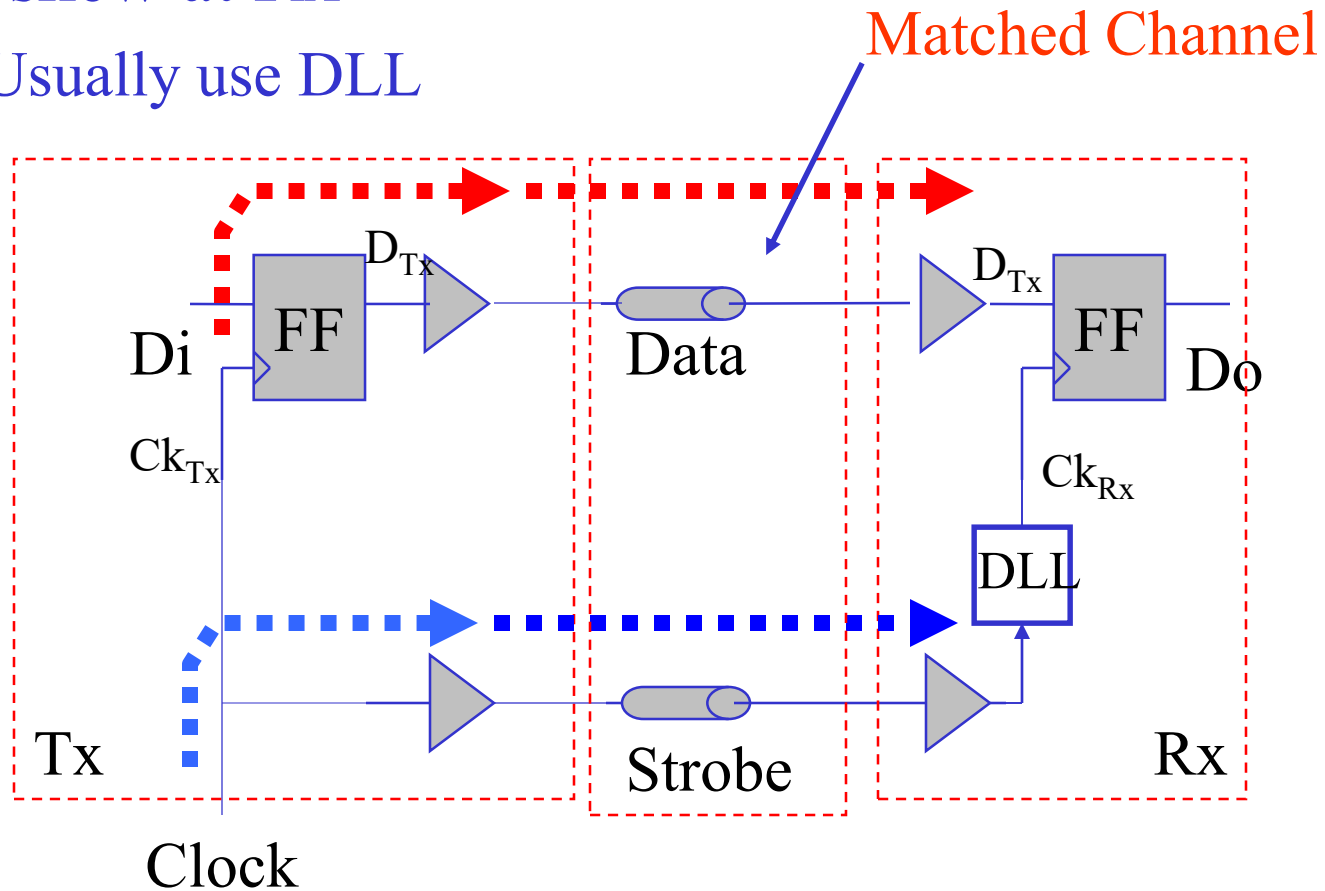
$$T > 2[\Delta(\delta_{DP} + \delta_{Tx} - \delta_{Rx} - \delta_{T/2}) + (t_{setup} + t_{hold})]$$

(100~1000) ps

Alternative Forwarded Clocking I/O (I)



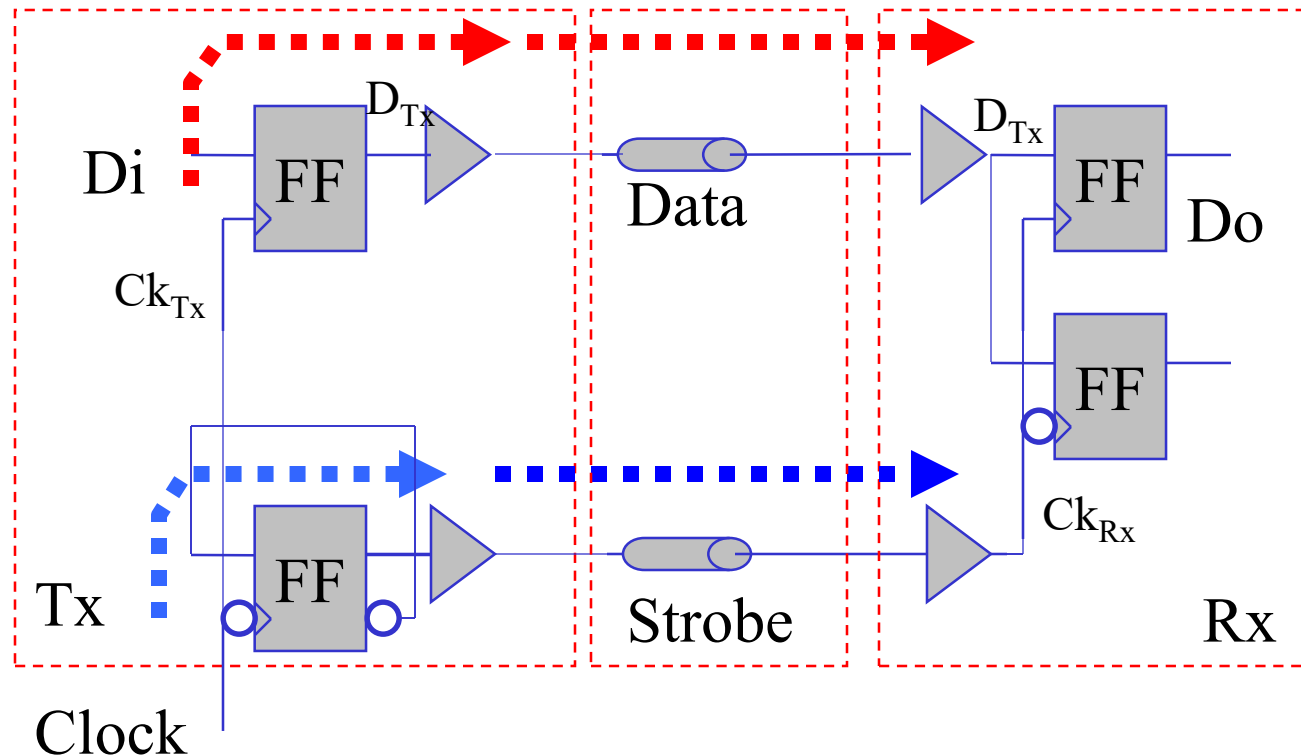
- De-skew at Rx
 - Usually use DLL



Alternative Forwarded Clocking I/O (II)



- Double-Data Rate
- Further Eliminated T_{cto} in timing equation



Embedded Clock I/O Architecture



- Embedded clock I/O is also known as the recovered clock I/O
 - Embedded clock scheme is the alternative solution to source sync I/O interfaces
 - Embedded clock I/O is a “adaptive delay” based clocking scheme
- Reduced clock distribution penalties
 - No dedicated clock channel requirement
 - Low cost
- High data rate
 - $>1\text{Gb/s}$
- Application examples: PCI-Express, USB2.0, Mipi Mphy, C-PHY interfaces

Embedded Clock I/O Architecture



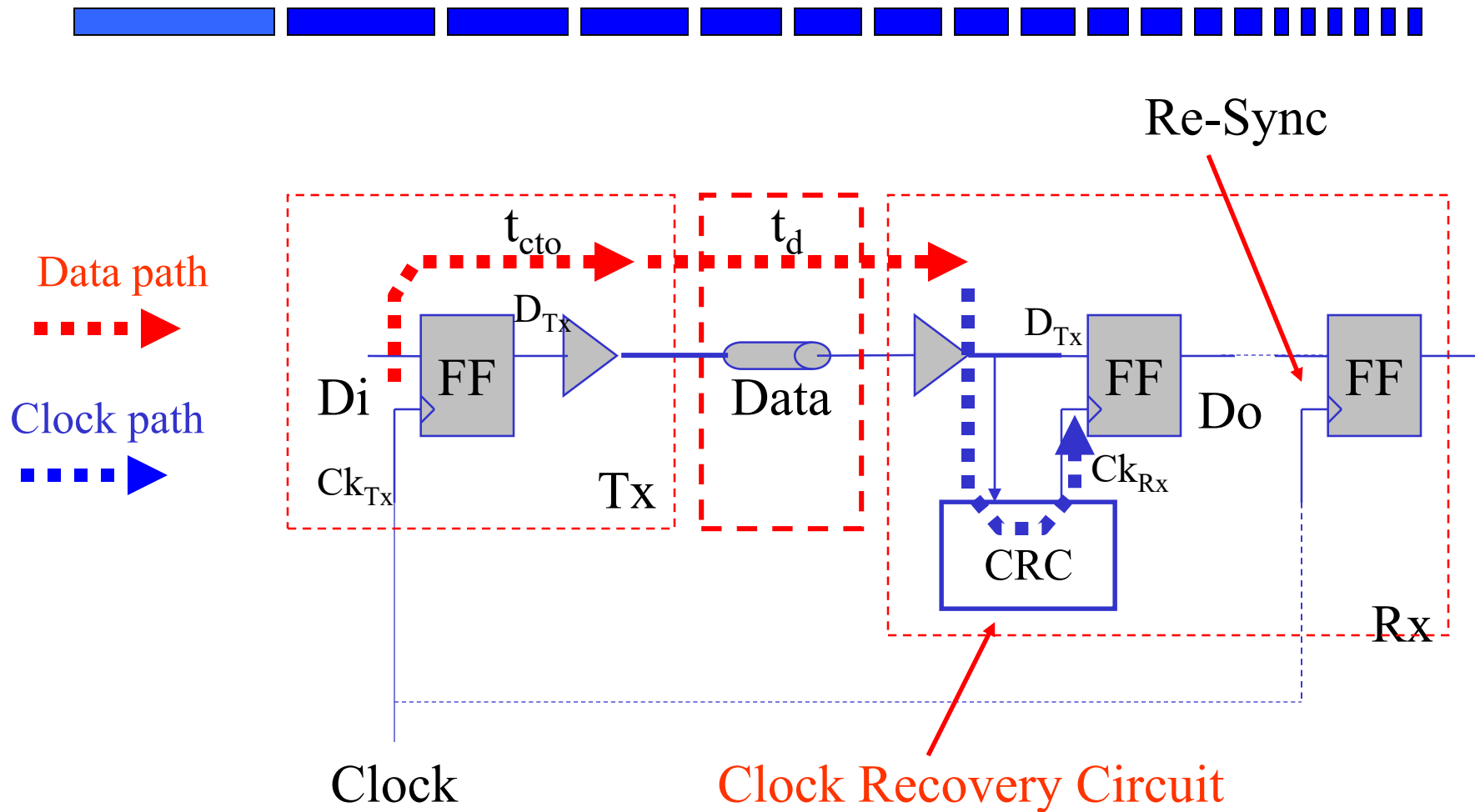
- Clock is extracted from received data in the embedded clock signaling
 - No strobe channel is required
 - Adaptive phase relation between Rx data and clock
- It is a further design concept shift from delay-matching based to adaptive feedback loop
 - Higher performance
 - Longer allowable signal/clock routing distance
 - Reduced signaling cost

Embedded Clock I/O Operation

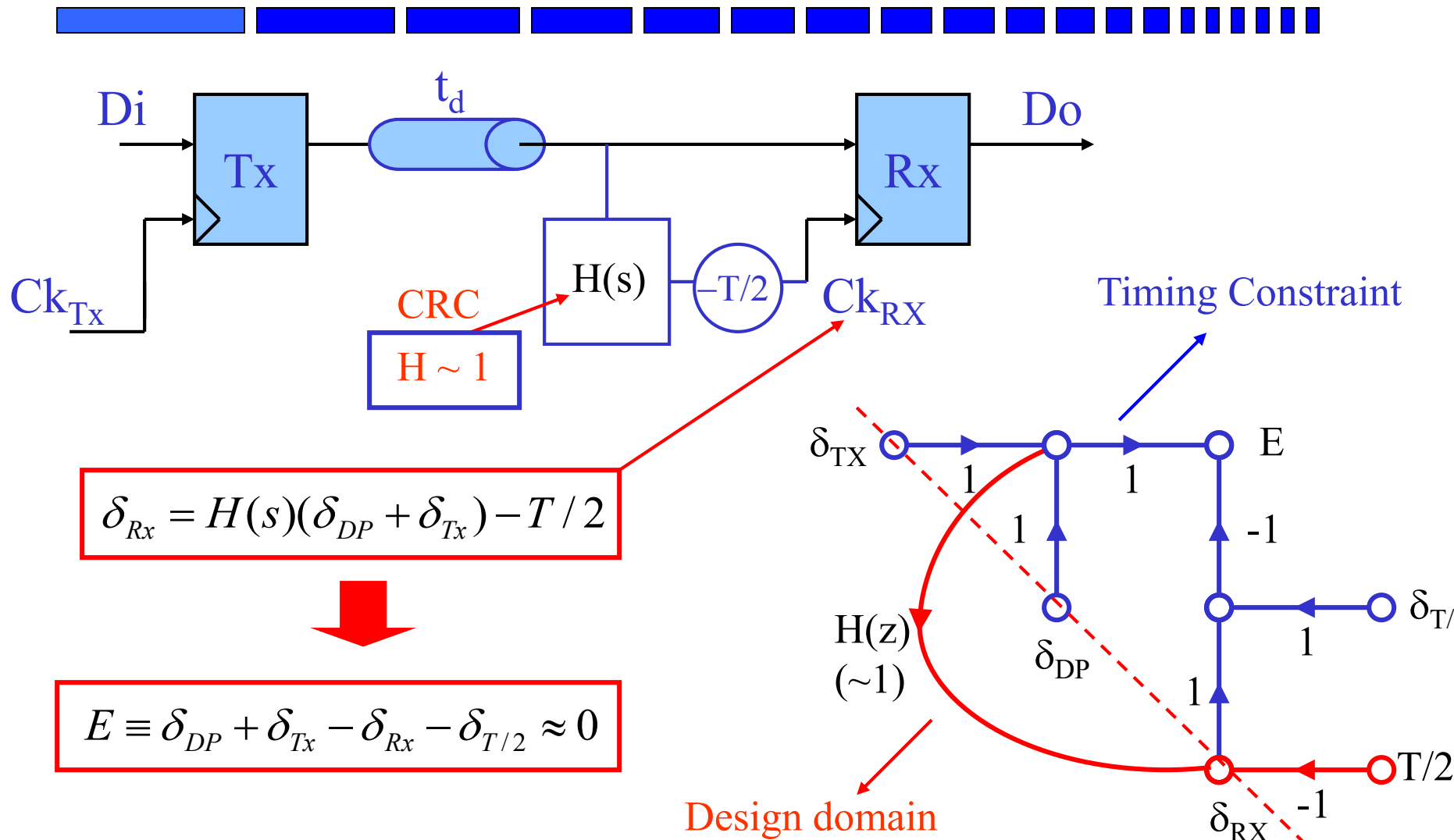


- Data D_{TX} generated/synched to the clock rise edges at the TX
- Data D_{TX} is propagated through the channel to generate D_{RX}
- Data D_{RX} is used to re-generate the Rx clock through the clock recovery circuit (CRC)
- Data D_{RX} is latched using the recovered Rx clock (data recovery circuit or DRC)
- Data D_o is re-synched to local Rx clock

Embedded Clocking I/O Operation



Embedded Clock I/O Circuits

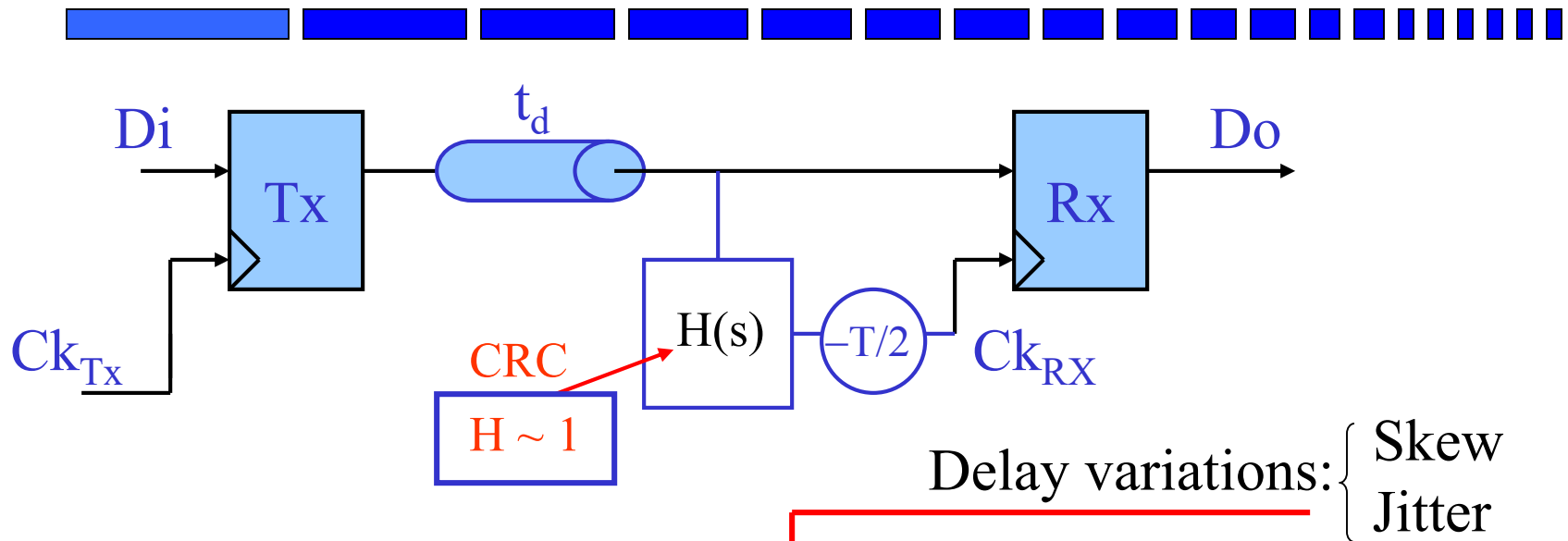


Common Sources of Clock Variations



- Loop bandwidth of the DRC loop
- Clock buffer jitter/skew

Embedded Clock I/O Circuits



$$E \equiv \Delta(\delta_{DP} + \delta_{Tx} - \delta_{Rx} - \delta_{T/2}) \leq E_{Max} \equiv \frac{T - (t_{setup} + t_{hold})}{2}$$

$$T > 2[\Delta(\delta_{DP} + \delta_{Tx} - \delta_{Rx} - \delta_{T/2}) + (t_{setup} + t_{hold})]$$

(100~1000) ps

Reading Assignment



- Please read Chapter 14 of the textbook

